Date	03 October 2026
Team ID	**SWTID-2026-7735**
Project Name	**Auto Ticket Classification using Flow Designer**
Maximum Marks	4 Marks

Project Design Phase-II

Technology Stack (Architecture & Stack)

Technical Architecture:

The deliverable includes the architectural diagram below and the specifications in Table 1 and Table 2.

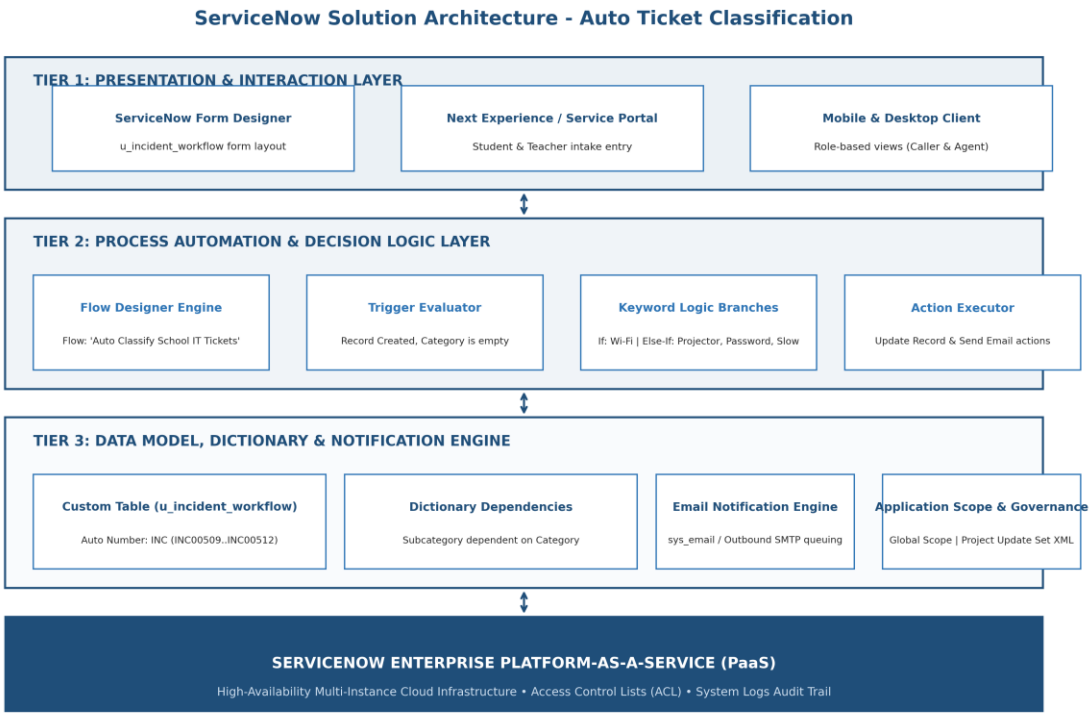


Figure 2.3: ServiceNow Enterprise Technology Stack & Tiered Architecture Diagram

Table-1 : Components & Technologies:

S.No	Component	Description	Technology
1	User Interface (UI)	Form interface for student and teacher ticket submission	ServiceNow Form Designer / Next Experience UI
2	Process Automation Engine	Evaluates creation triggers and executes flow logic	ServiceNow Flow Designer (Process Automation)

3	Decision Logic Engine	Parses keywords in Short Description and updates records	Flow Designer Natural Condition Builder & Data Pills
4	Database & Schema	Relational database storing incident records and metadata	ServiceNow MariaDB/MySQL Relational Database
5	Custom Data Model	Structured entity storing school IT tickets with Auto-Numbering	Custom Table: u_incident_workflow (Global Scope)
6	Dictionary & Taxonomy	Field relationships and hierarchical dependent choices	ServiceNow System Dictionary (sys_dictionary & sys_choice)
7	Notification Engine	Generates and sends automated email receipts to callers	ServiceNow Email Engine (sys_email) & SMTP Server
8	Access Control & Security	Controls table read, write, and create permissions	ServiceNow Access Control Lists (ACLs) & RBAC
9	Application Scope	Execution sandbox and deployment scope	Global Application Scope
10	Deployment & Migration	Captures all customization records for transport	ServiceNow Local Update Sets & XML Exporter
11	Cloud Infrastructure	Enterprise multi-instance cloud hosting platform	ServiceNow Cloud Datacenter Infrastructure

Table-2: Application Characteristics:

S.No	Characteristics	Description & Technology Implementation
1	No-Code Platform Framework	The solution completely avoids hardcoded JavaScript business rules, using Flow Designer visual logic.
2	Security & Governance Implementation	Enforces Role-Based Access Control (RBAC). Update Set tracking ensures clean XML migration.
3	Scalable & Reliable Architecture	Asynchronous trigger processing handles sudden bursts of ticket submissions during peak school hours.